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15 March 1976

MILITARY SPECIFICATION

TRANSDUCER, MOTIONAL PICKUP TR-299/G( )

This specification is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes requirements for Transducer, Motional Pickup TR-299/G( ), part of Alarm Set, Anti-Intrusion Restricted Area AN/GSS-26A, hereafter referred to as the "transducer".

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein. In the event of conflict between the requirements in the documents referenced herein and the contents of this specification the contents of this specification shall be considered a superseding requirement.

SPECIFICATIONS

MILITARY

MIL-E-4158	Electronic Equipment Ground: General Requirements For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements For Electronic Equipment

FSC 6350

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MIL-STD-810	Environmental Test Methods

HANDBOOKS

MILITARY

MIL-HDBK-217	Reliability Prediction of Electronic Equipment

DRAWINGS

ROME AIR DEVELOPMENT CENTER

Dwg 7518619 (Air Force Code Identification 07877)	Transducer, Motional Pickup TR-299/G( )

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Preproduction sample. When specified in the contract or purchase order, sample quantities of preproduction items shall be subjected to the tests specified herein. (See 6.2.1.b).

3.2 Production drawings. The transducer shall be fabricated and assembled in accordance with the drawings and documents listed on Drawing 7518619. (See 6.2.1.c).

3.2.1 General specification. Those designs, parts, materials, processes, identification markings and label requirements not covered by the drawings shall be in accordance with MIL-E-4158.

3.3 Characteristics.

3.3.1 Performance.

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3.3.1.1 General. The transducer is a magnetic and seismic/pressure sensitive cable suitable for burial under the surface of the earth where it detects magnetic and non-magnetic disturbances caused by an intruder. The transducer shall be capable of detecting the movement of nearby objects while providing minimum response to far-field disturbances.

3.3.1.2 Magnetic sensitivity. The transducer shall give a 1.00 ±0.05 millivolt (mV) peak-to-peak output when tested as specified in 4.8.2 when the center of the transducer section being tested is subjected to the peak magnetic field specified in table I:

TABLE I. MAGNETIC FIELD STRENGTH.

Frequency	Field (gauss peak)
1	0.0444 ± 0.0124
3	0.0171 ± 0.0042
10	0.00514 ± 0.00124
30	0.00196 ± 0.00032
45	0.00162 ± 0.00020

3.3.1.3 Sensitivity.

3.3.1.3.1 Core permeability. The relative permeability of the core sample at 50 ± 1 Hz when calculated from measurements made in accordance with 4.8.3 shall have a value not less than 1000.

3.3.1.3.2 Operational sensitivity. The sample mean value of detection probability of a set of three test transducers when tested in accordance with 4.8.4 shall not be significantly less than the sample mean value of a set of three reference transducers tested concurrently. The sample mean value of false and nuisance alarm rate for the test set of transducers shall not be significantly greater than the sample mean value of the reference set tested concurrently. A one-tailed significance test at the 0.05 level shall be used to determine whether significant differences exist between test and reference transducer groups.

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3.3.1.4 Transpositions. The transducer shall have 91 magnetic transpositions along its length at intervals of 1092 ± 25.4 millimeters (mm).

3.3.1.5 Resistance. The direct current (dc) resistance of the transducer shall be 265 ± 20 ohms at 25° ± 3°C.

3.3.1.6 Insulation resistance. Insulation resistance of the transducer from shield to sensor winding shall be 100 megohms minimum when tested at 500 ± 50 volts dc (Vdc).

3.3.1.7 Shielding. The transducer windings and insulation shall be covered by silver-copper braid shielding providing 85 percent or better coverage.

3.3.1.8 Connector interface.

3.3.1.8.1 Strain relief. The transducer shall be capable of withstanding a minimum pull force of 445 Newtons on the connector with the transducer assembly held 1422 mm from the connector end of the cable without electrical or mechanical damage.

3.3.1.8.2 Flexibility. The flexible end (1168 mm) of the transducer shall withstand 25 cycles of coiling around a 63.5 mm diameter form without electrical or mechanical damage.

3.3.2 Physical characteristics.

3.3.2.1 Weight. The weight of the transducer shall be 31.75 kilograms (kg) maximum.

3.3.2.2 Length. The length of the transducer shall be 102.7 +0.0 -0.9 meters (m). This length includes a right angle connector and 1168 ±50.8 mm of flexible interface cable.

3.3.2.3 Jacket. The transducer shall be provided with a continuous outer jacket providing moisture, rodent and abrasion resistance suitable for direct burial in the ground.

3.4 Quantitative reliability values. The transducer shall be designed to achieve a service life of 8 years.

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3.4.1 Analysis. The reliability of the transducer shall be predicted based on the stresses experienced by the parts using failure information contained in MIL-HDBK-217 Section 2, Part Stress Analysis Prediction. No other source of part failure rates shall be used unless specifically approved by the procuring activity.

3.5 Environmental requirements. (Unless otherwise specified, these requirements apply for the nonoperational, operational, transport and storage modes and shall be as specified in Section 4.) The transducer shall be designed and constructed to meet any individual or probable combination of service condition demands specified herein, without mechanical or electrical damage or performance degradation:

a. Temperature: When exposed to these low and high temperature extremes -54°C to 71°C. The transducer shall be capable of withstanding 163°C for a maximum of ten (10) minutes during installation in a hard surface when a hot pour sealant is used to encapsulate the transducer.

b. Low Pressure: When exposed to these low pressures:

(1) Non-operation, transit, storage: 140.6 mm of mercury (Hg) or 12.19 kilometers (km) above sea level.

(2) Operation: 446.5 mm of Hg or 4.27 km above sea level.

c. Leakage: When submerged in water to a nominal depth of 6.09 m no evidence of leakage shall occur.

d. Shock: After encountering shocks during operation, handling, transportation and service use.

e. Vibration: After encountering vibrations during operation and shipment by land, sea and air.

f. Humidity: When exposed to these low and high humidity extremes:

(1) Non-operating and storage: Up to 100 percent, including condensation due to temperature changes.

(2) Up to 100 percent with a temperature of 26.7°C.

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g. Fungus: When stored under warm, humid conditions conducive to fungus growth (30°C, 95 percent relative humidity). A certified statement of use of non-nutrient materials shall satisfy this requirement.

h. Salt Fog: When exposed to salt fog atmosphere as encountered in sea locations, coastal regions and during ocean transportation.

3.6 Transportability. The transducer shall not suffer any degradation when subjected to normal transportation methods while packed on its reel as specified in 5.1.1.

3.7 Nameplates and product marking. Product identification shall be marked near the connector end of the transducer. Marking shall be applied in a contrasting color. The transducer shall be marked after each pair of transpositions to show the locations for cutting.

3.8 Workmanship. The transducer shall be free of defects such as cracks, splits, voids, scratches, bends, kinks, and marking deficiencies sufficient to degrade performance or application. It shall be clean and free of foreign matter. Workmanship shall conform to Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. The tests defined herein are intended to verify that the transducer meets or exceeds the requirements of this specification.

4.1.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. (See 6.2.1.d).

4.1.1.1 Terms and definitions. Quality assurance terms and definitions shall be in accordance with MIL-STD-109.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Component and material inspection. (See 4.4)

b. Mechanical inspection. (See 4.5)

c. Preproduction inspection. (See 4.6)

d. Quality conformance inspection. (See 4.7)

4.3 Test conditions.

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests shall be conducted at 25°C ±5°C and local ambient pressure.

4.3.2 Test items. Unless otherwise specified, all tests shall be conducted on items which have been manufactured using the same drawings, materials, facilities, processes and procedures as will be used in regular production of the transducer.

4.4 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards. Previous acceptance or approval of material by the Government shall in no case be construed as a guarantee of the acceptance of the finished product.

4.5 Mechanical inspection. The transducer shall be given a thorough mechanical and visual inspection and test to determine that the quality of all materials and workmanship is in compliance with the requirements of this specification. Particular attention shall be given to the following:

		Requirement Paragraph
a.	Completeness	3.2
b.	Nameplates, identification markings and labels	3.2 and 3.7
c.	Finishes	3.2
d.	Fit of components in their respective positions	3.2
e.	Lubrication and rust prevention	3.2
f.	Fastening and security devices and parts	3.2

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g.	Weight check	3.3.2.1
h.	Overall dimension checks	3.3.2.2
i.	Jacket	3.3.2.3
j.	Workmanship	3.8
k.	Potential shorts and defective insulation	3.8
l.	Other visual defects	3.8

4.6 Preproduction inspection.

4.6.1 Preproduction sample. At the beginning of regular production, a sample quantity as specified in the contract (see 6.2.1.b) shall be subjected to preproduction tests to demonstrate the adequacy and suitability of the contractor's processes and procedures to achieving the performance inherent in the design. The sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production of the transducer.

4.6.2 Examination and test. The test samples shall be examined for conformance to 4.5 prior to conducting the preproduction tests. The preproduction tests shall consist of the tests identified by an "X" in Column 1 of table II.

4.6.3 Test failure. Failure of a test sample to comply with the preproduction inspection shall result in rejection of the sample. In the event of rejection, corrective action shall be taken and a new test sample (or a portion thereof subject to the prior approval of the Government) shall be resubmitted for test. (See 6.2.1.e).

4.7 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests (See 4.7.1)

b. Sampling tests (See 4.7.2).

4.7.1 Individual tests. Each transducer shall be examined for conformance to 4.5 and shall be subjected to the specific tests identified by an "X" in Column 2 of table II. Failure of any test shall be cause for rejection.

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4.7.2 Sampling tests. Transducers selected in accordance with 4.7.2.2 shall be subjected to the tests identified by an "X" in Column 3 of table II with an AQL level of 4.0.

4.7.2.1 Lot. A lot shall be defined as a quantity of transducers scheduled for delivery at the same time.

4.7.2.2 Lot sampling. Samples for tests shall be submitted in accordance with MIL-STD-105, inspection level S-1.

4.8 Test methods and procedures

4.8.1 Environmental tests. The transducer shall meet all the specification requirements when subjected to the following sequence of environmental tests in accordance with MIL-STD-810 with exceptions and deletions that follow. Performance tests of 4.8.6 and 4.8.7 shall be performed after each environmental test.

a. Temperature

(1) Low temperature: Method 502.1. Procedure I, Delete Steps 2 and 3.

(2) High temperature: Method 501.1. Procedure II, Delete Step 7 and replace with "Raise internal chamber temperature to 163°C until temperature stabilization of the transducer is reached. Maintain at this temperature for 10 minutes. Lower the internal chamber temperature to 71°C and maintain until temperature stabilization of the transducer is reached." In Step 8 the transducer shall be operated for a minimum of one hour.

b. Low pressure: Method 500.1, Step 2 change, "429.1 mm of Hg (16.9 inches of Hg or 15,000 feet above sea level)" to "446.5 mm of Hg (17.6 inches of Hg or 14,000 feet above sea level".

c. Leakage

(1) Test fixture: A test fixture suggested for performing leakage tests is shown in Figure 1 and contains the following elements:

(a) A hermetically sealed tank with removable lid, large enough to accommodate several transducers, a removable rack for supporting the transducers, and sufficient water to cover the transducers to a

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TABLE II. TEST SCHEDULE

Preproduction inspection	Quality conformance inspection		Test	Requirement paragraph	Test Method
	Individual	Sampling			
1	2	3	4	5	6
X			Reliability	3.4 and 3.4.1	Certification
X			Environmental tests	3.5	4.8.1
X			Low temperature	3.5.a	4.8.1.a(1)
X		X	High temperature	3.5.a	4.8.1.a(2)
X		X	Low pressure	3.5.b	4.8.1.b
X		X	Leakage	3.5.c	4.8.1.c
X			Shock	3.5.d	4.8.1.d
X			Vibration	3.5.e	4.8.1.e
X			Temperature/ humidity	3.5.f	4.8.1.f
X			Fungus	3.5.g	4.8.1.g
X			Salt Fog	3.5.h	4.8.1.h
X		X	Magnetic sensitivity	3.3.1.2	4.8.2
		X	Core permeability	3.3.1.3.1	4.8.3
X			Operational sensitivity	3.3.1.3.2	4.8.4
X		X	Transpositions	3.3.1.4	4.8.5
X	X		Resistance	3.3.1.5	4.8.6
X	X		Insulation resistance	3.3.1.6	4.8.7
X		X	Connector interface	3.3.1.8.1 and 3.3.1.8.2	4.8.8
X	X		Inspection of preparation for delivery	3.6 and 5.1 through 5.1.3	4.8.9

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depth of 152.4 mm or greater. The tank should be provided with viewing ports or similar means to render the submerged transducers fully visible from outside the tank. The tank should be provided with an exhaust port in the air space above the water.

(b) A vacuum system capable of lowering and maintaining air pressure in the tank at 140.6 mm of Hg or less. The system should contain an accurate gauge calibrated in millimeters of Hg or equivalent and a valve or other means to maintain lowered pressure in the tank.

(2) Leakage test: The leakage test shall be performed in accordance with the following procedure:

(a) Secure protective cap on transducer connector of each test article.

(b) Place units to be tested on the rack in the test tank.

(c) Lower the rack into the water, secure the lid, and turn on the vacuum pump making certain that the vacuum valve is open.

(d) When the gauge reads 140.6 mm of Hg or less, close the vacuum valve and turn off the vacuum pump. Maintain pressure in the tank of 140.6 mm of Hg or less throughout the remainder of the test.

(e) Observe units for evidence of gross leaks (indicated by the presence of a steady stream of air bubbles rising to the surface of the water). If this occurs, the units have failed the test and it should be immediately terminated by proceeding to (g). In this case performance tests of 4.8.6 and 4.8.7 shall not be performed.

(f) If no evidence of gross leaks appear, allow units to remain submerged for 15 + 5 minutes and observe for evidence of leaks: any formation of air bubbles emanating from within a unit and rising to the surface of the water.

(g) At the end of observation period, release vacuum, open tank, and raise platform out of water. Record any failures identifying the unit that failed.

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d. Shock: Method 516.2, Procedure I, terminal peak sawtooth pulse, 40g peak value, 18 milliseconds time duration, Figure 516.2-1.

e. Vibration: Method 514.2 Procedure X, Table 514.2-VII, Figure 514.2-7, Curve AX. Procedure XI Part 2.

f. Humidity: Method 507.1. Procedure II. Test measurements to be conducted between the 24th and 28th and between the 43rd and 48th hours of each cycle.

g. Fungus: Method 508.1, Procedure I, not required if requirement in 3.5g is furnished.

h. Salt fog: Method 509.1, Procedure I.

4.8.1.1 Acceptance criteria. A failure as a result of the environmental/mechanical tests shall be:

a. A condition in which the performance or physical configuration of the transducer does not meet the requirements of the functional tests;

b. A change in tolerance limits in any internal or external components which could in any manner prevent the transducer from meeting the operational requirements;

c. Any deterioration, corrosion or physical change which in the opinion of the cognizant Government agency, impairs the mechanical integrity of the transducer; and

d. Catastrophic or structural failure; and

e. Mechanical binding or loose parts that result in component failure or a hazard to personnel safety.

4.8.2 Magnetic sensitivity. This test shall be conducted on a complete transducer, instrumented as shown in Figure 2 and tested in accordance with 4.8.2.1. At each frequency, as specified in table I, the transducer shall be subjected to a magnetic field that produces a 1.0 $\pm$0.05 mV peak-to-peak output when applied to the center of the transducer section being

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tested. The averaged value of the field applied at the test coil center for the sample shall be as specified in table I. Sensitivity data shall be an average of ten readings taken at randomly selected points no closer than ten meters to either end. The test requirements do not include filter attenuation factors. Data shall be corrected for characteristics of filter being used at the test frequencies.

4.8.2.1 Test procedures.

a. Connect the equipment as shown in Figure 2.

b. Set the Hewlett-Packard 8875A or equivalent amplifier to a gain of 1000.

c. Position the transducer section to be tested in the center of the test coil within ±50 mm lengthwise; within ±25 mm radially. The transposition marking may be used to indicate the end of the section.

d. Set the signal generator to desired frequency.

e. Increase the signal generator output until amplified transducer output indicated on the oscilloscope is 1.0 ±0.05 volt peak-to-peak.

f. Measure the voltage across test coil and relate this voltage to magnetic field strength at the center of the coil. The magnetic field strength shall meet the requirement of 3.3.1.2.

4.8.2.2 Accept-reject criteria. A defective transducer shall be identified by failure to meet the requirements of 3.3.1.2 which shall be cause for rejection.

4.8.3 Core permeability test. This test shall be conducted on 1.25 meter long samples of stranded transducer core after wrapping with insulating tape but before solenoid windings are added. The instrumentation is shown in Figure 3. The ends of the test coil shall be located at least 1 meter from any objects containing ferromagnetic material. The relative permeability of the core sample at 50 Hz ±1 Hz shall meet the requirements of 3.3.1.3.1.

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4.8.3.1 Test procedures

a. Connect the equipment as shown in Figure 3. The magnetic core shall be removed from the test coil and be placed at least one meter from the ends of the coil.

b. Set the bridge control to "A.C. external," the parameter selector to "LP" and the scale selector to a range appropriate for measurement of the inductance of the air-core test coil. Adjust the detector sensitivity to a point approximately midway between the extremes of adjustment.

c. Set the signal generator for 50 $\pm$ 1 Hz, and adjust the output level to produce a substantial deflection of the bridge meter.

d. Adjust the digital and DQ arms of the bridge to obtain the deepest possible null. The value of inductance L_0 obtained from the bridge measurement shall contain not fewer than three significant figures. Record the measured value and the estimated uncertainty of the measurement.

e. Position the core section to be tested in the center of the coil within $\pm$ 50 mm lengthwise; within $\pm$ 10 percent of the coil diameter radially.

f. Adjust the bridge to obtain the best possible value of inductance L_1 of the test coil containing the core section. The value obtained shall contain not fewer than three significant figures. Record the measured value and the estimated uncertainty of the measurement.

g. Obtain the contribution L' of the core section under test to the measured inductance L_1 using the relationship:

$$L' = L_1 - L_0$$

The value L' shall contain not fewer than two significant figures. Record the value of L' and its uncertainty (sum of uncertainties of measurements of L_1 and L_0).

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h. Calculate the relative permeability μ' of the core sample at 50 Hz using the relationship:

$$\mu' = \frac{L'l}{\mu_0 S' N^2}$$

where l = length of test coil, meters

S' = cross-sectional area of core wires, m^2

N = number of turns on test coil

μ_0 = 4×10^{-7} henries per meter

L' = calculated value plus and minus measurement uncertainty, henries

The calculated relative permeability at 50 Hz of the core sample shall meet the requirements of 3.3.1.3.1 within the uncertainty of the measurement.

4.8.3.2 Accept-reject criteria. A defective transducer shall be identified by failure to meet the requirement of 3.3.1.3.1 which shall be cause for rejection.

4.8.4 Operational sensitivity. This test shall be performed on three complete transducers, made from the same materials and processed as a regular production run. Acceptance of the test items shall be based on a performance comparison with a Government-furnished sample of three acceptable transducers. Detection performance against magnetically clean human "intruders" and false and nuisance alarm performance over a cumulative test period of 100 hours shall be recorded for test and reference items. (See 6.2.1.f).

4.8.4.1 Test conditions.

4.8.4.1.1 Test and reference transducers. Test and reference transducers shall be installed using identical procedures. Transducers shall be tested for winding and insulation resistance before and immediately after installation. Any transducer found defective as a result of these tests shall be replaced. A settling time of two weeks following installation shall be allowed before beginning detection and false alarm tests.

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4.8.4.1.2 Intruders. Intruders for detection tests shall weigh not less than 50.43 kg and not more than 90.72 kg. They shall be clothed and equipped so as to be free of ferromagnetic material, including shoe nails, eyelets, arch supports, buckles, and zippers. Detection tests shall not be conducted when the following environmental conditions are present: noticeable precipitation, frozen ground, natural or man-made seismic disturbances, or winds with gusts exceeding 32 kilometers per hour (km/h).

4.8.4.2 Test procedures.

4.8.4.2.1 Detection test.

a. Connect the equipment as shown in Figure 4. Confirm that each channel is operational by self-test, crossing by a walking human, or other appropriate method. Have a magnetically clean intruder traverse the test field as shown in Figure 5 at a normal walking pace (nominal rate of 1.5 meters per second). The intruder shall cross all transducers perpendicularly, remaining at least 1.5 m from the last transducer crossed for a period of not less than 3 seconds. The next crossing shall be made in the opposite direction, approximately 3 meters along the transducer line from the first crossing. After crossing the last transducer and remaining at least 1.5 m beyond it for a period of not less than 3 seconds, the crossing pattern shall be repeated, with the intruder advancing approximately 3 m toward the end of the transducers after each crossing. Upon advancing to within 3 meters of the ends of the transducers, the intruder shall return along approximately the same path to the starting point. The complete crossing pattern shall then be repeated, so that each transducer is crossed a minimum of 120 times.

b. Record for each transducer the total number of intruder crossings attempted and detected.

c. Calculate the detection probability for each transducer and the sample mean detection probability and variance for the set of test transducers and for the set of reference transducers.

d. Test the sample mean values of detection probability for significant differences between test and reference transducers. The detection probability of test transducer shall meet the requirement of 3.3.1.3.2.

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4.8.4.2.2 False and nuisance alarm test.

a. Connect the equipment as shown in Figure 4. Confirm that each channel is operational by self-test, transducer crossing by a walking human, or other appropriate method. With no vehicles or personnel in the test field, the test and reference sensors shall be monitored for false and nuisance alarms for a cumulative period of not less than 100 hours. An observer shall be present during the test who shall attempt to locate sources of nuisance alarms. Any source of nuisance alarms that does not provide an alarm stimulus to test and reference transducers alike shall be noted (an example of such a nuisance alarm source would be an animal that is observed to cross some transducers but not others). Nuisance alarms due to sources providing unequal stimuli shall be disregarded in determining false and nuisance alarm rate. Alarms for which no cause can be observed shall be regarded as false alarms.

b. Determine sample means values of false alarm rate and nuisance alarm rate in alarms per hour for test and reference transducer sets. Include an explanation of any nuisance alarms disregarded per 4.8.4.2.2.

c. Test values of false and nuisance alarm rates for significant differences between tests and reference groups. False and nuisance alarm rates for test transducers shall meet the requirement of 3.3.1.3.2.

4.8.4.3 Accept-reject-criteria. A defective transducer shall be identified by failure to meet the requirement of 3.3.1.3.2 which shall be cause for rejection.

4.8.5 Transpositions. The transposition test shall be conducted with the transducer instrumented as shown in Figure 6 and tested in accordance with 4.8.5.1. The length of any transposition within the transducer is measured with a small test coil. With a given fixed excitation (level and frequency) applied to pins A and B of the transducer, the coil is moved along the transducer and the output is observed in the earphones. At each transposition, a maximum sound will appear in the phones. The distance between two successive maxima is the length of a transposition.

4.8.5.1 Test procedures.

a. Apply a full-wave rectified alternating current (AC), 120 Hz voltage to pins A and B of the transducer.

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b. Run the small test coil along the cable and listen for the maximum sound in the earphones to locate the transposition. The test coil should be held approximately 0.635 to 1.27 mm from the transducer.

c. Measure distance between transpositions and count the number of transpositions.

4.8.5.2 Accept-reject criteria. A defective transducer shall be identified by failure to meet the requirements of 3.3.1.4 which shall be cause for rejection.

4.8.6 Resistance. The resistance test shall be conducted with the transducer instrumented as shown in Figure 7. The dc resistance from A to B of the connector shall be measured.

4.8.6.1 Accept-reject criteria. A defective transducer shall be identified by failure to meet the requirements of 3.3.1.5 which shall be cause for rejection.

4.8.7 Insulation resistance. The insulation resistance test shall be conducted with the transducer instrumented as shown in Figure 8. The insulation resistance from A to C of the connector shall be measured with a megohmeter or equivalent at 500 volts $\pm$ 50 Vdc.

4.8.7.1 Accept-reject criteria. A defective transducer shall be identified by failure to meet the requirements of 3.3.1.6 which shall be cause for rejection.

4.8.8 Connector interface. The connector interface shall meet the specification requirements when subjected to the following tests. Performance tests of 4.8.6 and 4.8.7 shall be performed after each connector interface test.

a. Strain relief: Apply a minimum force of 455 Newtons on the connector with the transducer assembly held 1422 mm from the connector end of the cable.

b. Flexibility. Coil the flexible end (1168 mm) of the cable helically around a 63.5 mm diameter form and immediately uncoil the flexible end. Repeat 24 times.

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4.8.8.1 Accept-reject criteria. A defective transducer shall be identified by failure to meet the requirements of 3.3.1.8.1 and 3.3.1.8.2 which shall be cause for rejection.

4.8.9 Inspection of preparation for delivery. Inspection of packing, marking, and preparation for delivery shall be performed to determine compliance with 5.1 through 5.1.3.

5. PACKAGING

5.1 General. Transducers shall be packed in accordance with the contractor best commercial practice to insure acceptance by common carrier for safe transportation.

5.1.1 Packing. Each transducer shall be wound, without twisting, on a wooden reel with 609.6 mm (minimum) inner diameter and 1066.8 mm (maximum) outer diameter. The connector end shall be wound on last so that the connector is accessible for testing and deployment. The connector shall be firmly fastened to the reel end to prevent movement in shipment.

5.1.2 Reel construction. Either 9.52 mm (minimum) exterior plywood or 19 mm (nominal dimension) boards may be used for the ends of the reel. The center of the reel shall be wood, lined with 6.35 mm (minimum) cardboard. The covering over the transducer shall be 6.35 mm (minimum) cardboard.

5.1.3 Marking for shipment. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The transducer covered by this specification is intended for use in perimeter intrusion detection systems.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Number of preproduction test samples to be tested (see 3.1 and 4.6.1).

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c. In the event it becomes necessary to change the drawings (see 3.2) in order to achieve the performance requirements of the specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved such changes shall be processed in accordance with the provisions of the changes clause of the contract.

d. Inspection responsibility, if other than specified (see 4.1.1).

e. Instructions for selection of a new test sample in the event of failure of the first test sample (see 4.6.3).

f. The location and conditions for the non-magnetic field test (See 4.8.4).

6.2.2 Contract data requirements. Data requirements in support of the Quality Assurance Provisions (Section 4), including test plans/procedures, analytical and inspection/test reports, and submission of detailed test data, should be listed directly on the Contract Data Requirements List (CDRL), (DD Form 1423) that is incorporated into the contract.

6.3 United States customary equivalents. For ease of readability of the specification the US customary equivalents for the metric values specified herein are listed below instead of in text.

LENGTH

millimeters	inches
0.635	0.025
1.27	0.050
6.35	0.250
9.52	0.375
19	0.750
25	1*
50	2*
63.5	2.5
140.6	5.54
152.4	6
446.6	17.60
609.6	24
1066.8	42
1092 $\pm$ 25.4	43 $\pm$ 1
1168 $\pm$ 50.8	46 $\pm$ 2
1422	56

*Approximate

meters	inches
0.141	5.551
0.282	11.102
0.423	16.653
0.564	22.204
0.705	27.755
0.846	33.307
0.987	38.858
1	39.37
1.1	43.307
1.128	44.409
1.250	49.212
1.269	49.960
1.410	55.511
1.5	59.055

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LENGTH (CONTINUED)

meters	feet
1.52	5
3	9.84
6.09	20
102.7 +0.0 -0.9	337 +0.0 -3.0

kilometers	feet
4.27	14,000
12.19	40,000

TEMPERATURE

Degrees Celsius (°C)	Degrees Fahrenheit (°F)
-54	-65
25 ± 3	77 ± 5
25 ± 5	77 ± 9
26.7	80
30	86
71	160
163	325

WEIGHT

kilograms	pounds
31.75	70
50.43	120
90.72	200

FORCE

445 Newtons	100 pound-force

VELOCITY

32 km/h	20 miles per hour

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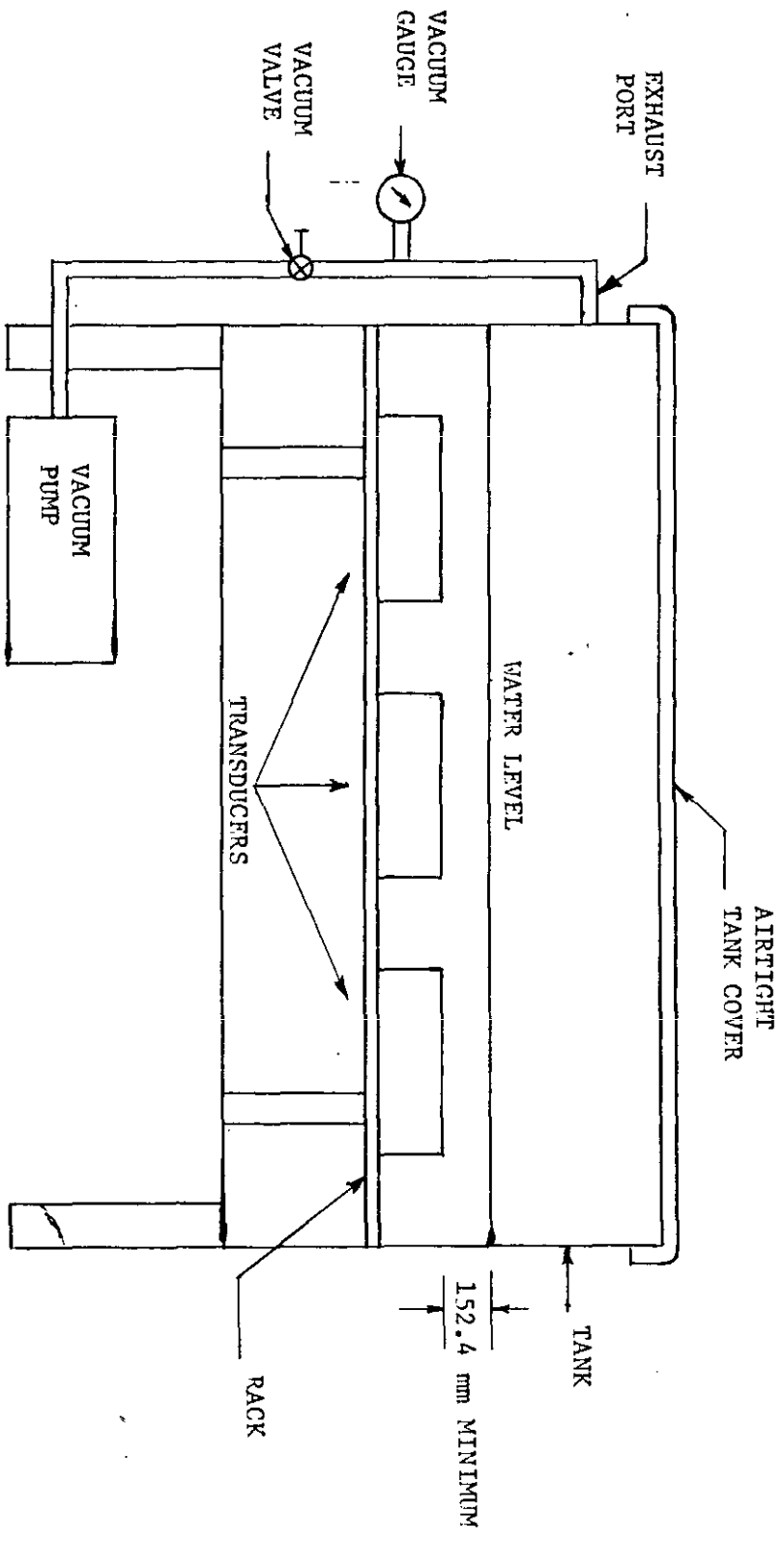
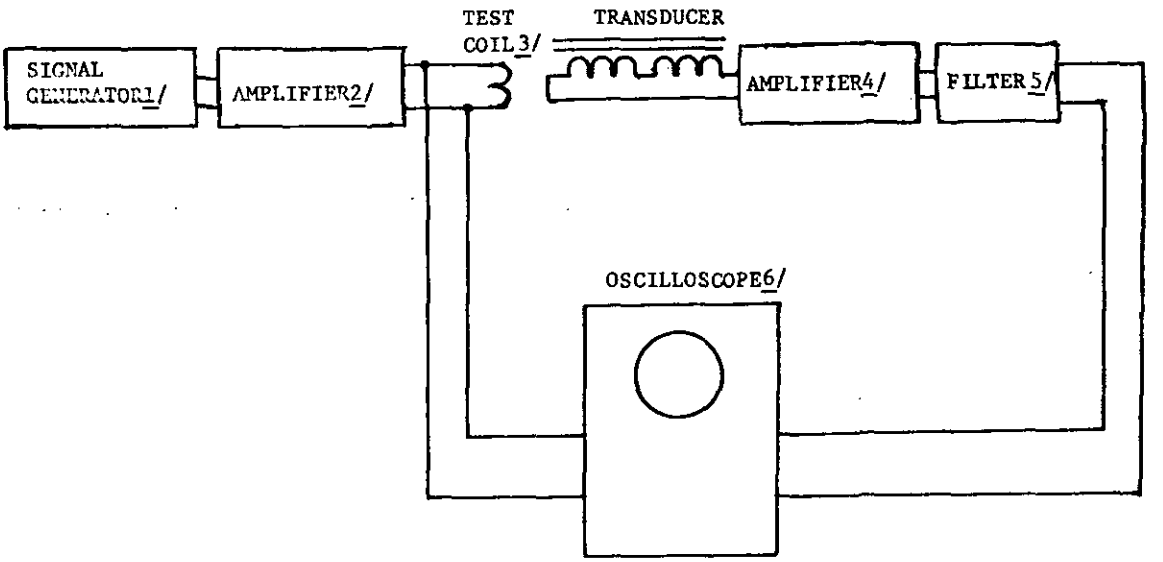


FIGURE 1. LEAKAGE TEST.

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1/ General Radio Type 1162A or equivalent.
Accuracy: ±1%

2/ Special purpose amplifier for amplification of output signal generator to the level needed by the test coil to develop required magnetic strength.

3/ The test coil shall be calibrated for magnetic field strength versus applied voltage and shall create a magnetic field strength relative to the center which varies in both directions according to the following:

METERS FROM CENTER	RELATIVE STRENGTH
0	1.000
0.141	0.994
0.282	0.973
0.423	0.921
0.564	0.789
0.705	0.520
0.846	0.250
0.987	0.117
1.128	0.062
1.269	0.037
1.410	0.024

Accuracy: ±5%

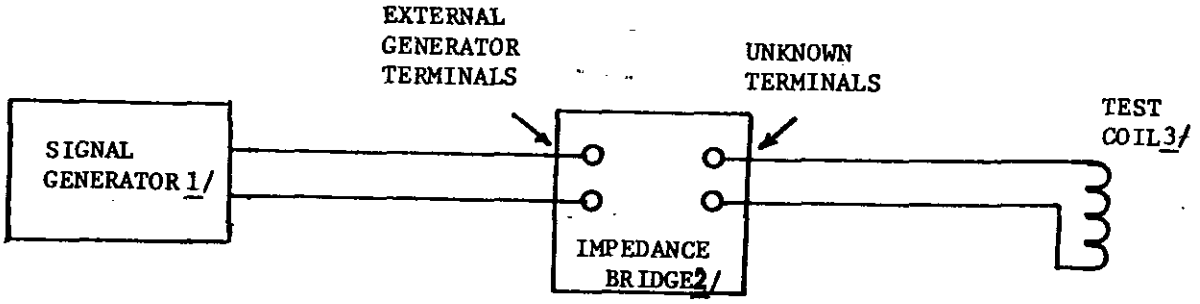
4/ Hewlett-Packard 8875A or equivalent.
Accuracy: ±0.1%

5/ Type: Low-Pass, 10th Order Chebyshev
1 dB passband ripple
F^{-3} = 50 Hz

6/ Tektronic Type 564 with 2 or more channel plug-in, or equivalent.
Accuracy: ±5%

FIGURE 2. MAGNETIC SENSITIVITY TEST.

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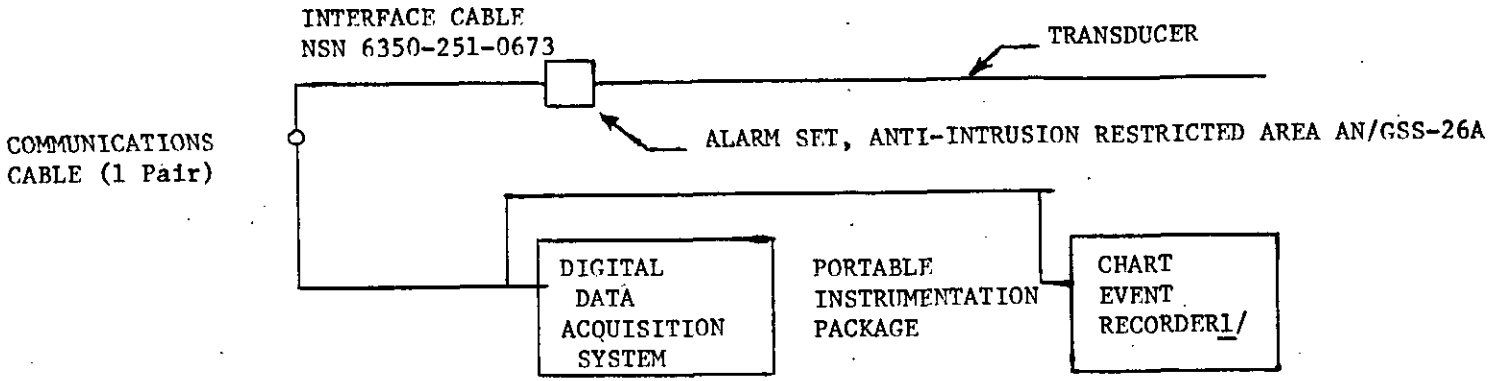
1/ General Radio Type 1162A or equivalent.
Accuracy: ±1%

2/ General Radio Type 1656 or equivalent.
Accuracy: ±1%

3/ The test coil used for the core permeability test shall be an air-core solenoid having a length of not less than 1.1 m, wound with not less than 500 turns of low-resistance wire. The Q factor of the coil at a frequency of 50 Hz shall not be less than 1.0. The test coil shall be equipped with non-ferromagnetic supports so that a sample of the transducer stranded core can be suspended along the axis of the coil. The test coil specified for the magnetic sensitivity test may be used for the core permeability test as well, provided it meets the requirements of this paragraph.

FIGURE 3. CORE PERMEABILITY TEST.

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1/ Brush Visicorder or equivalent.

FIGURE 4. EQUIPMENT CONNECTION (ONE OF SIX CHANNELS).

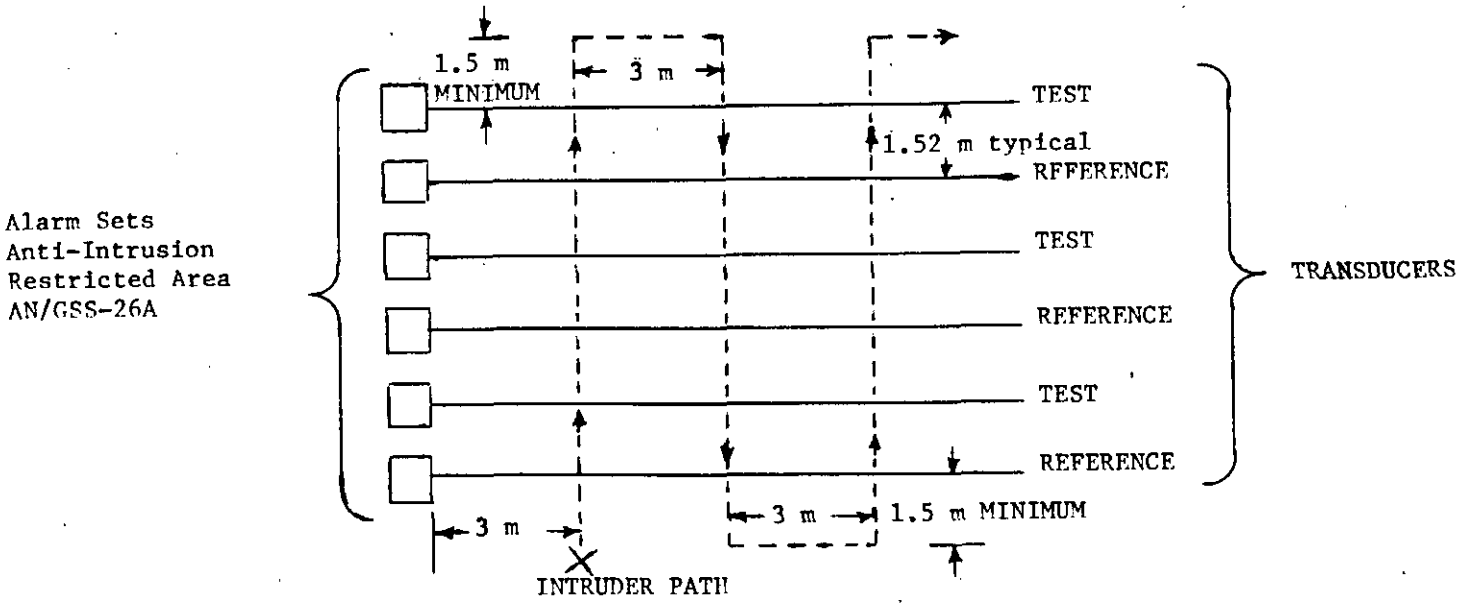


FIGURE 5. PLAN VIEW OF TEST FIELD SITE.

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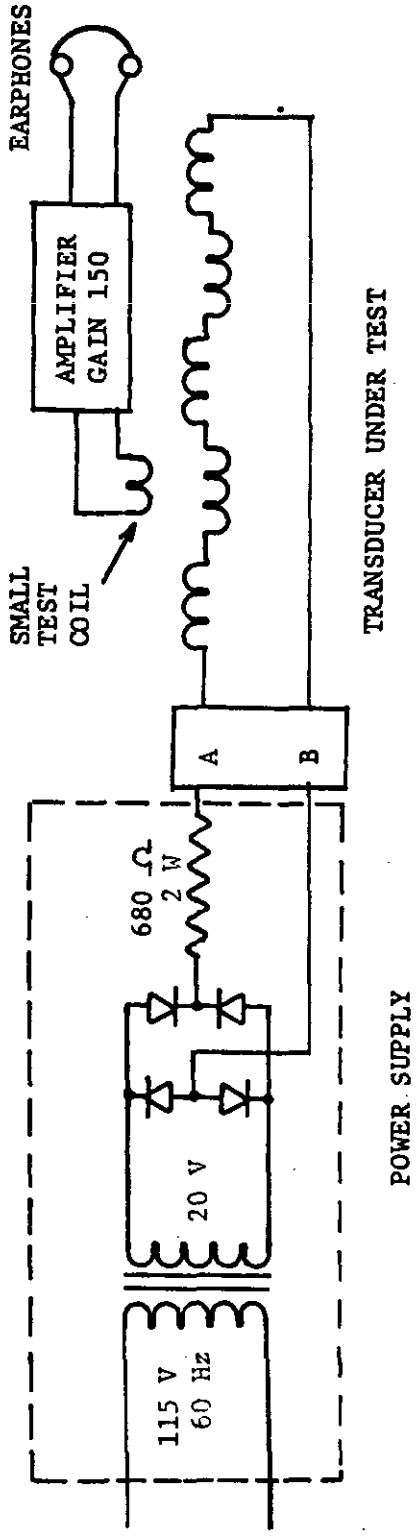
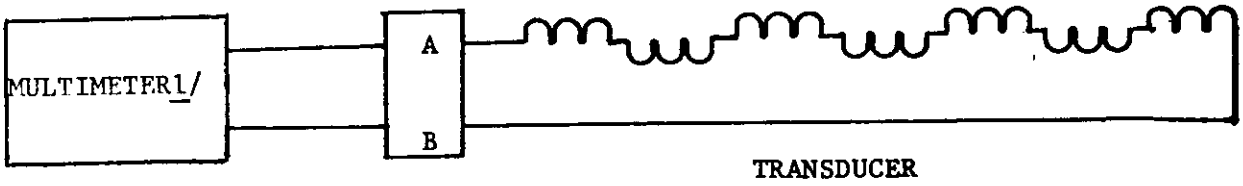


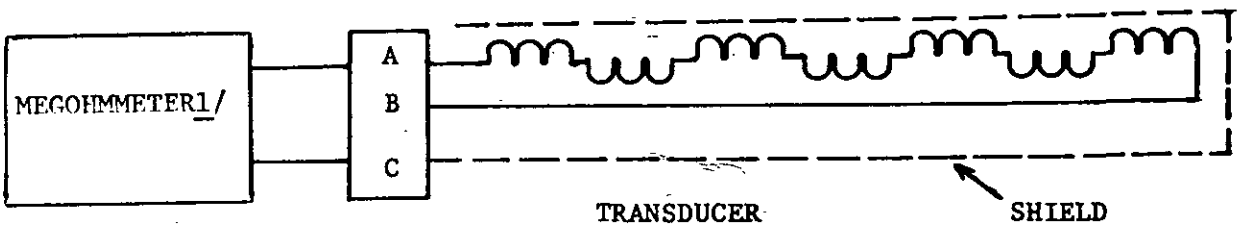
FIGURE 6. TRANSPOSITION TEST.

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1/ Fluke Model 8000 or equivalent.
Accuracy: ±1%

FIGURE 7. RESISTANCE TEST.



1/ General Radio Model 1862 or equivalent.
Accuracy: ±3%

FIGURE 8. INSULATION TEST.

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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